



**Verified Carbon
Standard**

MALAYSIA RENEWABLE BIOENERGY GROUPED PROJECT



CarbonTrace
Creating a Net Zero Future



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1 PROJECT DETAILS

1.1 Summary Description of the Project

Grouped Project

The MALAYSIA RENEWABLE BIOENERGY GROUPED PROJECT (hereinafter referred to as "the Project") is a grouped project which consists of projects supplying renewable electricity to the grid that are located in Malaysia. The project is owned by CarbonTrace Sdn.Bhd. (hereinafter referred to as the Project Owner), and the implementation of the projects is by the individual plant operators of each project activity instance (PAI). The purpose of this Grouped Project is to generate renewable electricity from sources such as biogas recovery from agricultural effluent, industrial effluent, food waste, and others, as well as biomass combustion from agro/forest biomass residues in waste streams that are sent to landfills. The electricity generated shall be supplied to the national electricity grid, via Tenaga Nasional Berhad.

Project Activity Instance 1 (PAI1) Description

The first PAI1 is located at Liziz Biogas Sdn Bhd in the state of Gua Musang, Kelantan, Malaysia. The biogas power plant is owned by Liziz Biogas Sdn Bhd as a subsidiary of Mestron Holdings Berhad, with technical guidance from True Eco Sdn Bhd and carbon strategy provided by CarbonTrace Sdn Bhd. The purpose of the Project is to generate electricity from the biogas recovered during the POME treatment process. The biogas in the absence of the project would be released into the atmosphere, thereby contributing to greenhouse gas (GHG) emissions.

The biogas power plant started exporting electricity to the Grid on 26th July 2023 by utilizing POME from Liziz Plantation Sdn. Bhd.'s palm oil mill as its feedstock. This mill was registered on July 19, 2000, and has been operational since then. The proposed project activity includes the installation of an anaerobic wastewater treatment facility that employs Complete Stirred Tank Reactor (CSTR) biogas reactor technology. The biogas recovered is expected to be used for:

- a. electricity generation for own consumption, and/or
- b. electricity generation to the national grid, with the option of flaring for any excess quantity generated.

In the first PAI1, GHG emission reductions are achieved by capturing biogas from POME treatment, which prevents its release into the atmosphere and substitutes grid electricity with renewable energy. Prior to PAI1's implementation, the baseline electricity needs were met through conventional means, which involved consuming non-renewable energy sources or energy from the national grid, primarily dependent on fossil fuels. POME treatment method would have involved a series of cascading open lagoon ponds. Thus, the baseline scenario represented a significant environmental impact due to GHG releases and reliance on non-renewable energy sources.

The methodology used in this Project is AMS-I. D: Grid connected renewable electricity generation, based on the renewable electricity generated from the Biogas Power Plant. The proposed project activity is expected to reduce 10249.20 tCO₂e annually or total of 71744.40 tCO₂e during the 7-years of the first crediting period.

1.2 Audit History

PAI1

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation</i>	26072023-26072030	VCS	ERM-CVS	Seven Years
<i>Verification</i>	26072023-26072024	VCS	ERM-CVS	One Year

1.3 Sectoral Scope and Project Type

Grouped Project

The PAIs in this Project shall involve the construction and operation of a new power plant/unit or retrofit, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant that uses renewable energy sources and supplies electricity to the grid.

The PAIs shall have electricity generation capacity of less than 15MW and will generate emission reductions of less than 60ktCO₂e per year. The type of project is thus in accordance with the limit of a small-scale project.

PAI1

PAI1 involves generation of renewable electricity via the recovery of fugitive methane emissions from Palm Oil Mill Effluent followed by the treatment of the methane/biogas captured.

Sectoral scope ¹	Energy Industries
Project activity type	Type I: Renewable Energy Projects Category I.D: Grid Connected Renewable Energy Generation

1.4 Project Eligibility

1.4.1 General eligibility

Grouped Project

The general eligibility for each PAI shall be detailed below.

PAI1

As per section 2.1.1 of VCS Standard (version 4.7), the scope of the VCS Program includes:

1. The seven Kyoto Protocol greenhouse gases: PAI1 is designed to capture the methane gas from wastewater generated in the Palm Oil Mill to generate renewable electricity. Methane is one (1) of the seven (7) Kyoto Protocol greenhouse gases, therefore PAI1 is applicable to this scope.

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

2. Ozone-depleting substances (ODS): Not applicable.
3. Project activities supported by a methodology approved under the VCS Program through the methodology development and review process: Not applicable.
4. Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions): The applied methodology AMS-I. D (Version 18.0) of PAI1 is a methodology approved under CDM Program, which is a VCS approved GHG program.
5. Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not applicable. PAI1 does not involve REDD+ programs and projects.

As per section 2.1.2 of VCS Standard (version 4.7), the scope of the VCS Program excludes the projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction: PAI1 involves the generation of renewable electricity which would have been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. Hence, it does not generate GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.

As per section 2.1.3 of VCS Standard (version 4.7), the VCS also excludes the following project activities under the circumstances indicated in Table 1: Excluded Project Activities. PAI1 involves the generation of renewable electricity which would have been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid, which does not fall under the exclusion listed in Table 1.

PAI1 starts in operation on 26/07/2023 and shall complete validation before 25/07/2025. As per VCS Standard (version 4.7), validation shall not begin until the 30-day public comment period has begun, and the validation/verification body shall not complete validation until after the 30-day public comment period has ended. Hence, the estimated time for completion of validation and verification is sufficient to meet the deadline of meeting the requirements of “*non-AFOLU projects shall complete validation within two years of the project start date*”.

The methodology implemented is AMS-I. D, which qualifies under the VCS Program. According to AMS-I. D (Version 18.0), the eligibility threshold for a small-scale Clean Development Mechanism

(CDM) project activity is set at 15 MW. PAI1 with an installed capacity of 2.76 MW and a net export capacity of 2.00 MW, meets the eligibility criteria. PAI1 is not a fragmented part of a larger project or activity that would otherwise exceed such limits, as there is **no other** project within one kilometer of the project boundary where:

1. The project proponents for both projects are the same.
2. The sectoral scope and project activity for both activities are the same.
3. The other projects have been registered under the VCS Program or other GHG program within the previous 2 years.

1.4.2 AFOLU project eligibility

Not applicable as Project is a non-AFOLU project.

1.4.3 Transfer project eligibility

Not applicable as Project is not transferred from other GHG Programs.

1.5 Project Design

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

Grouped Project Design

Eligibility criteria for the inclusion of new project activity instances

The project is a grouped project, and the following are the eligibility criteria for the inclusion of new project activity instances into the grouped project activity:

- **Applicability Conditions:** The project activity instances shall meet applicability conditions for applicable methodology of AMS I.D version 18.0, as defined in Section 3.2.
- **Geographical Area:** The Project activity instances are located in the geographic boundary for the grouped project, that is, Malaysia.

- **Baseline Scenario:** All Project Activity Instances shall meet the baseline scenario set by the methodology of AMS I.D version 18.0, as set out in section 3.4.
- **Technology type:** Project Activity Instances to be included in the grouped project is the technology involving power generation from renewable sources such as biogas recovery from agricultural effluent, industrial effluent, food waste, and others, as well as biomass combustion from agro/forest biomass residues in waste streams that are sent to landfills.
- **Additionality:** The project activity instances to be added as part of the grouped project will meet the additionality criteria as set out in the methodology. As per AMS I.D (Version 18.0) Clause 13, the additionality shall be proven by applying the “General guidelines for SSC CDM methodologies, information on additionality and “General guidance on leakage in biomass project activities”.
- **Start Date:** The start date of each project activity instance under the grouped project shall not be prior to the start date of the grouped project. The start date of each project activity instances shall be proven through official documentation.

Compliance of Project Activity Instance 1 (PAI1)

No.	Eligibility criteria for inclusion	Compliance of Project Activity Instance 1
1.	Applicability Conditions: The Project Activity Instances shall meet applicability conditions for applicable methodology of AMS I.D version 18.0, as defined in Section 3.2.	PAI 1 is in compliance with the applicability conditions for applicable methodology of AMS I.D version 18.0, as defined in Section 3.2.
2.	Geographical Area: The Project Activity Instances are located in the geographic boundary for the grouped project, that is, Malaysia.	PAI 1 is located in Malaysia, with detailed location as explained in Section 1.13.
3.	Baseline Scenario: All Project Activity Instances shall meet the baseline scenario as set out in the methodology of AMS I.D version 18.0, as explained in section 3.4.	PAI 1 meet the baseline scenario as set out in the methodology of AMS I.D version 18.0, as explained in section 3.4.
4.	Technology type: The Project Activity Instances to be included in the grouped project is the technology involving power	PAI 1 involves the renewable electricity generation from biogas. The biogas is

	generation from renewable sources such as biogas recovery from agricultural effluent, industrial effluent, food waste, and others, as well as biomass combustion from agro/forest biomass residues in waste streams that are sent to landfills.	captured from Palm Oil Mill Effluent and treated to produce biogas.
5.	Additionality: The Project Activity Instances to be added as part of the grouped project will meet the additionality criteria as set out in the methodology. As per AMS I.D (Version 18.0) Clause 13, the additionality shall be proven by applying the “General guidelines for SSC CDM methodologies, information on additionality and “General guidance on leakage in biomass project activities”.	PAI 1 meets the additionality criteria as set out in the methodology, following the “General guidelines for SSC CDM methodologies, information on additionality and “General guidance on leakage in biomass project activities”.
6.	Start Date: The start date of each Project Activity Instance under the grouped project shall not be prior to the start date of the grouped project. The start date of each project activity instances shall be proven through official documentation.	The start date of the grouped project is 26-July-2023, which is the date of commercial operation of biogas plant for PAI1. Hence, this condition is fulfilled.

1.6 Project Proponent

PAI1

Organization name	CarbonTrace Sdn Bhd
Contact person	Cliff Siaw Sze Hau
Title	Director
Address	D2-1-05, 1, Jalan Solaris Dutamas 1, Solaris Dutamas, 50480 Kuala Lumpur, Malaysia
Telephone	+6016-858-9345
Email	cliff@carbon-trace.com

Organization name	True Eco Sdn Bhd
Contact person	Wong Yee Yien, Elaine
Title	Technical Consultant
Address	No. 2a-1 & 2a-2, Jalan Equine 10b, Taman Equine, 43300 Seri Kembangan, Selangor, Malaysia
Telephone	+6011-63301266
Email	elainewong@true-eco.com

1.7 Other Entities Involved in the Project

PAI1

Organization name	Liziz Biogas Sdn Bhd
Role in the project	Powerplant Operator for PAI1
Contact person	Por Teong Eng
Title	Director
Address	PT 50102, Jalan MU 1/9, Seksyen 10, Taman Perindustrian Meranti Utama, Puchong 47120 Selangor, Malaysia
Telephone	+ 03-8069 1815
Email	por@mestron.com.my

1.8 Ownership

Grouped Project

CarbonTrace Sdn Bhd is the sole project owner of the Project. The ownership of every additional PAI to be added to this Grouped Project will be provided through Carbon Service Agreement between CarbonTrace and the powerplant owner.

PAI1

The project proponent has entered into Carbon Service Agreement with the biogas plant owner, Liziz Biogas Sdn. Bhd, which has the legal rights to operate PAI 1. The evidence of the right of use which can be referred to in the carbon service agreement, operating license and Letter of Award for Feed-in Tariff issued by the Sustainable Energy Development Authority Malaysia (SEDA Malaysia).

1.9 Project Start Date

Grouped Project

Project start date	26- July-2023
Justification	As per section 3.8 of VCS Standard (version 4.7), the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals. In PAI1, the project began generating GHG emission reductions on 26th July 2023, which can be proven in the Letter of Award for Feed-in Tariff issued by the Sustainable Energy Development Authority Malaysia (SEDA Malaysia).

1.10 Project Crediting Period

Grouped Project

As per the VCS Standard (version 4.7) the total project crediting period for non-AFOLU projects shall be either seven years (twice renewable for a total of up to 21 years) or ten years fixed

PAI1

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	26-July-2023 to 25-July-2030

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/ removals of PAI1 are:

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

For the first (renewable) crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
26-July-2023 to 31-December-2023	4464.72
01-January-2024 to 31-December-2024	10249.20
01-January-2025 to 31-December-2025	10249.20
01-January-2026 to 31-December-2026	10249.20
01-January-2027 to 31-December-2027	10249.20
01-January-2028 to 31-December-2028	10249.20
01-January-2029 to 31-December-2029	10249.20
01-January-2030 to 26-July -2030	5784.48

Total estimated ERRs during the first or fixed crediting period	71744.40
Total number of years	7
Average annual ERRs	10249.20

1.12 Description of the Project Activity

Grouped Project

The Grouped Project shall consist of multiple PAIs for the implementation of renewable electricity generation that is derived from bioenergy, such as biogas recovery from agricultural effluent, industrial effluent, food waste, and others, as well as biomass combustion from agro/forest biomass residues in waste streams that are sent to landfills. In the baseline scenario, electricity would have been generated by the operation of grid-connected power plants.

PAI1

The PAI1 is implemented at the 2MW biogas powerplant owned by Liziz Biogas Sdn Bhd, which is a **greenfield** powerplant at site location next to a palm oil mill. This PAI captures biogas from the wastewater generated by the mill and utilizes it to generate electricity to the powerplant and to the national grid. There are 2 gas engines included in the project boundary, with a capacity of 1.2 MWe and 1.56 MWe each.

Prior to the implementation of project, the baseline wastewater treatment facility would be a series of cascading open lagoon ponds to treat the POME and the baseline scenario for electricity would be the generation of electricity by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Technology Description

The generation of biogas from Palm Oil Mill Effluent (POME) involves a series of steps designed to efficiently convert organic waste into renewable energy. This process can be structured into several key components, including cooling pond, pretreatment system, anaerobic reaction system, biogas treatment, biogas power generation, POME treatment and biogas flaring system.

The POME is first channeled to the cooling pond to reduce the temperature of the effluent to optimal levels for anaerobic reactions. Anaerobic bacteria, responsible for breaking down organic matter in POME to produce biogas, operate efficiently within a specific temperature range. If POME is too hot, it can inhibit these bacteria, decreasing the efficiency of biogas production. Thus, cooling ponds are essential for ensuring that the temperature of POME is adjusted to levels conducive to maximizing biogas yield and, subsequently, electricity generation.

The cooled POME is then sent for pretreatment system to prepare it for efficient biogas production. Pretreatment includes removing solids and adjusting pH levels, making it suitable for anaerobic digestion.

The pre-treated effluent is then fed into anaerobic reaction tanks, which are the Continuous Stirred Tank Reactor (CSTR) consisting of 2 units with a capacity of 6200m³ for each unit. The tanks used are Glass-Fused-to-Steel (GFtS) tanks, which are durable with a lifespan of up to 50 years and resistant to the corrosive nature of POME. The CSTR is continuously stirred to ensure homogeneity, optimizing the contact between the microorganisms and the substrate. In this oxygen-free environment, microorganisms break down the organic matter in the POME, producing biogas—a mixture of methane (CH₄), carbon dioxide (CO₂), and other gases.

Following its generation, biogas contains hydrogen sulfide (H₂S). Due to the presence of a large amount of water vapor in the biogas, the water and H₂S in the biogas act together to accelerate the corrosion and blockage of metal pipelines, valves, and flow meters. In addition, SO₂ produced by the combustion of H₂S combines with water vapor in the combustion products to form sulphurous acid, which corrodes the metal surface of equipment and pollutes the atmosphere, affecting health of plant operators. Therefore, the biogas desulfurization system is in place to remove H₂S.

The biogas is then passed through a dehumidification system to remove moisture. This step is critical to prevent condensation and corrosion in the biogas utilization equipment. The dehumidification system in PAI1 reduces the moisture content of biogas by lowering its temperature to its dew point temperature and causing the water vapor in the biogas to condense out.

The desulphurized and dehumidified biogas is then conveyed to two generation set, with a capacity of 1.2 MWe and 1.56 MWe each. The biogas serves as a fuel to power the engine, which in turn drives the generator to produce electricity. The electricity is then passed through the transformer system, before being supplied to the powerplant and to be injected into the national grid electricity system. Management of leakage is done with a gas tester by service engineer where any excess biogas not utilized for electricity generation is safely combusted in a biogas flaring system. The flaring system can be operated on auto mode, where the system will start once detected a set pressure figure. All parameters necessary for the operation and management of the biogas are integrated to an integrated system to be checked by service engineer.

1.13 Project Location

Geographic area for Grouped Project

Project Activity Instances shall be within the border of Malaysia, Geodetic coordinates are provided in a KML file and uploaded in Verra.

Geographic area for PAI1

Physical Address: Ladang Kuala Betis
Pt No 5893 Mukim Ulu Nenggeri
Daerah Bertam, Jajahan Gua Musang
18300 Gua Musang, Kelantan Darul Naim,
Malaysia

Coordinates : 101.8058942° E 4.8727409° N

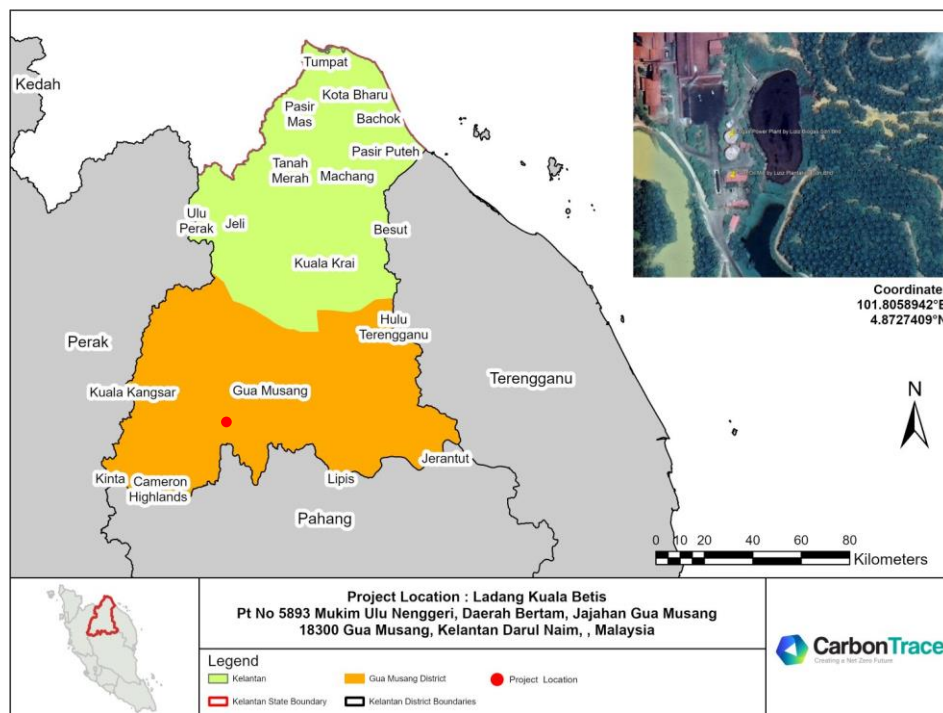


Figure 1: Project Location

1.14 Conditions Prior to Project Initiation

Grouped Project

The conditions prior to project initiation for each PAI shall be explained below and detailed in Section 3.4.

PAI1

PAI 1 activity is implemented at a Greenfield location. The electricity generated by the biogas powerplant to the grid would have been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

PAI 1 has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal, and destruction. This is demonstrated as the project emission is zero, as per stated in AMS-I.D “*For most renewable energy project activities, project emissions =0.*” with exclusion for geothermal powerplants and hydro powerplants. Since PAI 1 involves a biogas powerplant, it does not generate emissions but only reduce the baseline emissions, which is the electricity which would have been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Moreover, the net emission reduction is calculated based on the net export capacity of the powerplant. As the baseline scenario is the same as the conditions existing prior to the project initiation, refer to Section 3.4 (Baseline Scenario).

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Grouped Project

Every additional PAI to be added to this Grouped Project is required to demonstrate compliance with all relevant local, regional and national laws, statutes and regulatory formworks.

PAI1

PAI1 is compliant with the conditions of local laws and regulations in Malaysia, including but not limited the following:

- Environmental Quality Regulations (Designated Premises) (Crude Palm Oil) 1977 – P.U.(A) 342/77 Regulation 6 - Making Changes that change the quality of the effluent.
- Environmental Quality (Clean Air) Regulations 2014
- Renewable Energy Act 2011 [Act 725]
- Sustainable Energy Development Authority Act 2011

PAI 1 has all the necessary documents including property deed, construction permit and operational permit. PAI1 has also successfully received the approval for Feed-In Tariff from the Sustainable Energy Development Authority (SEDA) Malaysia.

In Malaysia, there is currently no specific regulation that mandates the management or use of biogas from palm oil mill projects. However, the Malaysian Palm Oil Board (MPOB²) has set guidelines that require all new palm oil mills built after January 1, 2014, to have wastewater treatment systems that include methane capture technology. This requirement also extends to existing mills that seek to increase their capacity after this date. However, there was an exemption introduced on May 22, 2021, for existing mills expanding their capacity to less than 270,000 metric tonnes of Fresh Fruit Bunches (FFB) per year.

Liziz Plantation Sdn Bhd, a palm oil mill established in 2000, falls outside these requirements because it was built before 2014. Despite this, the mill is planning for future expansions that will push its productivity over the 270,000 MT FFB per year limit. Consequently, although the mill could benefit from emission reductions by managing wastewater under the AMS-III.H methodology, this project does not consider such benefits. Instead, the project focuses on the AMS-I.D methodology, which pertains to Grid Connected Renewable Electricity Generation. Neither the MPOB nor the Malaysian government mandates the use of biogas for electricity generation, meaning Liziz Plantation Sdn Bhd is not obligated to utilize biogas in this manner.

² https://e-lesen.mpob.gov.my/document/CRITERIA%20AND%20GUIDELINES%20ON%20MPOB%20LICENSE%20APPLICATION_EnglishVersion_latest.pdf

1.16 Double Counting and Participation under Other GHG Programs

Grouped Project

Every additional PAI to be added to this Grouped Project will indicate whether the PAI is involved in double counting and participation under other GHG programs.

PAI 1

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

Grouped Project

Every additional PAI to be added to this Grouped Project will indicate whether the PAI is involved in double claiming, other forms of credit and Scope 3 emissions.

PAI 1

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes

☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

Grouped Project

Every additional PAI to be added to this Grouped Project shall indicate its sustainable development contributions.

PAI 1

PAI1 utilizes cutting-edge biogas technology to convert Palm Oil Mill Effluent (POME), a significant byproduct of palm oil production, into renewable energy. Located in a rural area, PAI1 captures methane from POME, which would otherwise be released into the atmosphere, and converts it into electricity. This electricity is supplied to the local grid, enhancing energy security and reducing reliance on fossil fuels. The initiative also provides substantial local benefits, including job creation and improved air quality, promoting renewable energy within the community.

The expected SD contributions are:

1. Affordable and Clean Energy (SDG 7):

By transforming POME into biogas and generating electricity, PAI1 increases the share of renewable energy in the national grid. This contributes to a more sustainable and accessible energy supply. The efficient conversion of waste into energy optimizes resource use in the palm oil industry, enhancing overall energy efficiency.

2. Economic Growth and Employment (SDG 8):

The construction and operation of the biogas plant create various employment opportunities for residents. These range from technical roles in engineering and plant operation to unskilled labour, supporting diverse economic growth. By providing these job opportunities, PAI1 stimulates local economies and supports sustainable development in rural areas.

3. Industry, Innovation, and Infrastructure (SDG 9):

PAI1 exemplifies the innovative application of biogas technology, showcasing how industrial waste can be repurposed for energy production. The successful implementation of this technology can inspire similar initiatives, promoting wider adoption of sustainable technologies.

4. Climate Action (SDG 13):

By capturing and utilizing methane from POME, PAI1 significantly reduces greenhouse gas emissions. Methane is a highly potent greenhouse gas, and its reduction plays a crucial role in mitigating climate change. This reduction directly supports global and national efforts to combat climate change and minimize environmental impacts.

Alignment with Malaysia's Sustainable Development Priorities

PAI 1 aligns closely with Malaysia's sustainable development goals. Malaysia aims to reduce carbon intensity by 45% by 2030 and increase renewable energy to 70% of the national energy mix by 2050. Additionally, Malaysia emphasizes economic development that also protects the environment. PAI 1 contributes to these national goals by increasing the use of renewable energy, reducing greenhouse gas emissions, and supporting economic growth in rural communities. To ensure the project's contributions are effectively tracked and reported, a comprehensive monitoring framework is established. This includes regular assessment of greenhouse gas emissions reductions, energy efficiency, and the socioeconomic

impact on local communities and annual reporting to stakeholders, such as the Malaysian Palm Oil Board (MPOB) and the Sustainable Energy Development Authority (SEDA), to demonstrate progress towards sustainable development goals and national priorities.

1.18.2 Sustainable Development Contributions Activity Monitoring

PAI 1

Brief Description

During the monitoring period of 26-July-2023 to 31-July-2024, PAI1 continued to capture methane emissions from Palm Oil Mill Effluent (POME) using advanced biogas technology, located in a rural area. The captured methane was converted into renewable electricity and supplied to the local grid, reducing greenhouse gas emissions and supporting clean energy initiatives. PAI1 contribute to Sustainable Development Goals (SDGs) as follows:

- **SDG 7 (Affordable and Clean Energy):** Increased the share of renewable energy by converting POME into electricity.
- **SDG 8 (Decent Work and Economic Growth):** Created jobs in rural areas through the operation of the biogas plant.
- **SDG 9 (Industry, Innovation, and Infrastructure):** Demonstrated innovative use of biogas technology for energy generation from waste.
- **SDG 13 (Climate Action):** Reduced greenhouse gas emissions by capturing methane.

PAI1 supports Malaysia's priorities to reduce carbon intensity, increase renewable energy usage, and promote economic growth while protecting the environment. It aligns with national goals for sustainable energy (SDG 7), economic development (SDG 8), industrial innovation (SDG 9), and climate mitigation (SDG 13). Monitoring and reporting on these contributions include regular assessments and annual reporting to stakeholders such as the Malaysian Palm Oil Board (MPOB) and the Sustainable Energy Development Authority (SEDA). Evidence of these contributions is provided in the appendices to this report.

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	Added 3574.2 MWh of renewable electricity to the local grid during the monitoring period	Contributed 3574.2 MWh of renewable electricity to the local grid since project start.
2)	8.5	8.5.2 Unemployment rate, by sex, age and persons with disabilities	Implemented activities to decrease	Created 20 new jobs during this period, enhancing local employment.	Created a total of 20 jobs since the project's inception.
3)	9.4	Utilization of biogas from POME to generate electricity	Implemented activities to increase	Captured and utilized 643,361 m ³ of biogas to be utilized for electricity generation.	Captured and utilized 643,361 m ³ of biogas to be utilized for electricity generation.
4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	Prevented the release of 2090.9 tonnes of CO ₂ -equivalent methane during the monitoring period.	Cumulatively avoided 2090.9 tonnes of CO ₂ -equivalent emissions over the project's lifetime.

1.19 Additional Information Relevant to the Project

Leakage Management

Grouped Project

As per AMS-I.D, Version 18.0, “General Guidance on Leakage in Biomass Project Activities” shall be followed. Based on Table 1: Emission source per type of biomass, in the General Guidance on Leakage in Biomass Project Activities, the source considered for “biomass residues or wastes collected and used” is the competing use of biomass.

PAI 1

Since PAI1 does not involve competing use of biomass, there will be no leakage to be considered

Commercially Sensitive Information

There is no commercially sensitive information that is or needs to be excluded from the public of the project description.

Further Information

Not applicable. There is no additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the GHG emission reductions or carbon dioxide removals, or the quantification of the project’s reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

PAI1

Stakeholder Identification	The process used to identify stakeholders likely to be impacted by PAI1 is by extending a radius of 10km from the PAI1 site at the coordinate of 101.8058942° E 4.8727409° N, as shown in the figure below. The potential list of stakeholders identified are: • Liziz Plantation Mill Workers • Kilang Sawit LPM workers • Communities residing in Kuala Betis • Communities residing in Kampung Langsat • Communities residing in Kampung Angkek As per VCS Standard (Version 4.7), distant or intermittent user groups who will be affected in very limited ways by the project need not be defined as stakeholders. The communities residing in Kuala Betis, Kampung Langsat and Kampung Angkek would be affected in very limited ways by the project as they are distant from project location and separated by mountainous terrains.
Legal or customary tenure/access rights	The legal ownership and land-use rights of the area for PAI1 belong to Liziz Plantation Sdn Bhd and Liziz Biogas Sdn Bhd. The agreement to develop the biogas powerplant and register as a VCS project can be referred to the Carbon Service Agreement between Liziz Biogas Sdn Bhd and CarbonTrace Sdn Bhd. As the area is a rural area, communities nearby only consist of plantation (rubber plantation and palm oil plantation) and mill workers. There are no conflicts regarding tenure or access rights, whether between different community groups, individuals, or with the state.

Stakeholder diversity and changes over time

Based on the data derive from WorldPop³ as shown in the figure below, it can be derived that the population density around Project area is extremely low. As such, there is a lack of information on population around PA11 area located in the subdistrict of Kuala Betis, other than the population size of 3800⁴ as of 2020. From 2010 to 2020, the population increased 53.5% from 2476 to 3800.

From a bigger picture, the demographic of Gua Musang is summarized as follow:

Table: Principal statistics of population on census year, Gua Musang, Kelantan⁵

Year	2000	2010	2020
Ethnic group of citizens			
Bumiputera	66,729	76,823	88,598
Chinese	3,718	3,870	4,758
Indians	263	350	834
Others	93	161	337
Percentage of ethnic group of citizens (%)			
Bumiputera	94.2	94.6	93.7
Chinese	5.3	4.8	5.0
Indians	0.4	0.4	0.9
Others	0.1	0.2	0.4

As shown in the table, the ethnic groups of citizens in Gua Musang, Kelantan in 2020 consist predominantly Bumiputera at 93.7%, which includes Malays and Orang Asli, followed by 5.0% of Chinese, 0.9% of Indians and 0.4% of "Others". The "Others" category encompasses individuals not falling into Malaysia's three primary ethnic groups: Malay, Chinese, or Indian. This group includes people from diverse backgrounds, such as Baba and Nyonya, Kadazandusun, Iban and Kashpalati, as well as workers from other countries employed in various sectors like agriculture and plantations.

The changes in the make-up of each group over time is minimal over the two decades for Bumiputera and Chinese, while changes are observed in

³ [Open Spatial Demographic Data and Research - WorldPop](#)

⁴ [City Population with Source from Department of Statistics Malaysia](#)

⁵ [Department Of Statistics Malaysia- 2020 Key Findings on Population and Housing Census of Gua Musang, Kelantan](#)

the Indian group and “Others” group from 0.4% and 0.1% in 2000 to 0.9% and 0.4% in 2020 respectively.

Table: Selected statistics household income and expenditure, Gua Musang, Kelantan

Year	2016	2019
Monthly statistics household gross income		
Median (RM)	2,818	2,942
Mean (RM)	3,551	4,009
Monthly households consumption expenditure		
Median (RM)	1,906	2,514
Mean (RM)	2,230	2,929
Gini Coefficient and incidence of poverty		
Gini Coefficient	0.340	0.367
Incidence of poverty	27.2	17.2

Based on the table⁶ above, there was an improvement in the economic conditions of Gua Musang's population between 2016 and 2019. Household gross income saw a rise, both median and mean, suggesting overall growth in earnings. Monthly consumption expenditure also increased, indicating possibly higher living standards or inflation effects. The Gini Coefficient's rise from 0.340 to 0.367 points towards growing income inequality. However, the significant decrease in poverty incidence from 27.2% to 17.2% marks substantial progress in reducing poverty.

Expected changes in well-being

The biogas power plant project in Gua Musang, by converting POME into electricity, is expected to improve local well-being and environment. It leads to cleaner energy sources, reduces greenhouse gas emissions, and enhances local air and water quality. PAI1 also provides job opportunities, boosting the local economy. This aligns with sustainable development goals, promoting environmental conservation and community well-being.

Location of stakeholders

The identified stakeholders located within a 10km radius of the project site are surrounded palm oil plantations and mills. Notably, the Kuala Betis community, situated 3.85km northwest of PAI1, is the closest community located.

⁶ Department Of Statistics Malaysia- 2020 Key Findings on Population and Housing Census of Gua Musang, Kelantan

Location of resources	The territories and resources owned by stakeholders include palm oil plantation and palm oil mills. There is no customary access lands within the area as the lands, territories and resources and issued via formal legal title to owners.
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2.1.2 Stakeholder Consultation and Ongoing Communication

PAI1

Date of stakeholder consultation	28-June-2024
Stakeholder engagement process	The announcement of meeting was done through the email and WhatsApp two weeks prior to the meeting date. The languages used to conduct the meeting were Malay and English. Attendance list was given to the attendees to record their attendance, The agenda of the meeting is as follows: 1. Registration 2. Welcoming Speech 3. Presentation on description of project activity, technology implemented and environmental and socio-economic benefits of the project activity. 4. Presentation on carbon credit & validation and verification process 5. Checking on their work safety & first aid 6. Feedback from the stakeholders, Q&A session 7. Conclusion The outcomes of the meeting are recorded in the meeting minutes.
Consultation outcome	All attendees are in favour of the implementation of PAI1. They are also keen to know more on how does PAI1 contribute to the bigger picture of reducing the effects of

	climate change. All the workers are safely secured by the Occupational Safety and Health Act 1994 and Factories and Machinery Act 1967. They are also well informed of the validation and verification process required from Verra and VVB.
Ongoing communication	Ongoing communication with stakeholders will be done through phone calls and emails.
Stakeholder input	There are no updates to the project design from the input received during the consultation

2.1.3 Free Prior and Informed Consent

PAI1

Obtaining consent	PAI1 does not affect any IPs, LCs, and customary rights holders. There are no ongoing or unresolved conflicts regarding the project activity.
Outcome of FPIC	There is no consent needed from the IPs, LCs and customary rights holders as the PAI1 does not involve any encroachment of land rights of other parties other than Project proponents and implementation parties. The project activity is constructed within the lots owned by Liziz Biogas Sdn Bhd.

2.1.4 Grievance Redress Procedure

PAI1

Development process	The development grievance redress procedure is done by the community liaison officer and discussed with the implementation partner. This is to ensure the process for receiving, hearing, responding, and attempting to resolve
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	grievances within a reasonable time is feasible and practical for PAI1 to be implemented smoothly.
Grievance redress procedure	The grievance redress procedure includes establishing accessible channels for stakeholders to submit grievances. The channels include hotline, email, and physical drop boxes in strategic locations. The community liaison officer will then ensure grievances are recorded and acknowledged. Upon receipt, each grievance would be assessed preliminary to categorize the issue's nature and urgency. The process would then involve a detailed investigation, respecting confidentiality, and impartiality, to understand the grievance's context and stakeholders' perspectives. Following the investigation, the aggrieved parties and relevant stakeholders will be engaged through meetings or mediation sessions to propose culturally sensitive conflict resolution methods. A resolution plan, including corrective actions and timelines would then be proposed while ensuring transparency and fairness throughout the process. The aggrieved parties would have the opportunity to provide feedback on the resolution plan before its implementation.

Grievances received	Resolution and outcome
No significant grievance raised	No significant grievance raised

2.1.5 Public Comments

PAI1

No comments received thus far.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

PAI1

The management teams have expertise and experience in implementing similar project activities. True Eco Sdn Bhd, the technical consultant has over 10 years of experience in the development and commercialisation of water and wastewater treatment, biogas generation and refining powerplant and environmental protection facilities including polishing plant, sludge processing system, and others.

2.2.2 Risk Assessment

PAI1

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	For natural risks, biogas facilities are generally constructed to withstand local weather conditions such as heavy rain and flood. PAI1 does not introduce significant changes to local ecosystems that could disrupt wildlife or natural habitats. It utilizes existing waste streams from palm oil production without altering the surrounding environment.
Risks to stakeholder participation	No risk identified.	The stakeholders participating in the PAI 1 are the biogas powerplant workers. Safety training and proper Personal Protective Equipment (PPE) are provided to the workers to ensure a safe working environment. Hence, the risk in this context is not applicable in this project.
Working conditions	No risk identified.	The safety of working conditions at the powerplant is in

		accordance with the Occupational Safety and Health Act 1994 of Malaysia. Emergency Plan and First Aid Management is available in the Biogas Powerplant. Therefore, this risk is not applicable to this project.
Safety of women and girls	No risk identified.	The organization works under the various rules and regulations of the host country. Hence the safety of women and girls are taken care of in the powerplant. Therefore, this risk is already mitigated.
Safety of minority and marginalized groups, including children	No risk identified.	The PAI1 is within the premises of the biogas powerplant. Therefore, there is no risk to the safety of minority and marginalized groups, including children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified.	PAI1 adheres to all the laws and regulations in Malaysia regarding the pollutants limit such as air, noise, discharges to water, generation of waste and release of hazardous materials.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

PAI1

	Risks identified ⁷	Mitigation or preventative measure(s) taken
Discrimination	No risk identified.	All personnel will be treated as per the law of land with no discriminations.

⁷ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Sexual harassment	No risk identified.	All personnel will be treated as per the law of land with no sexual harassment.
Equal pay for equal work	No risk identified.	All personnel will be treated as per the labour law of Malaysia with equal pay for equal work.
Gender equity in labor and work	No risk identified.	To ensure equal opportunities in terms of gender equity and pay for labor and work, PAI1 has adopted comprehensive strategies. These include implementing transparent hiring practices, offering equitable pay scales based on roles and responsibilities rather than gender, and providing training and advancement opportunities equally to all employees. Additionally, efforts to maintain a balanced representation of genders in all job categories and decision-making positions are prioritized. Monitoring mechanisms are in place to regularly assess and adjust these practices, ensuring ongoing commitment to equality and fairness in the workplace.
Forced labor	No risk identified.	The plant will be operated as per the law of the country, forced labor is strictly prohibited in the laws of Malaysia.
Child labor	No risk identified.	The plant will be operated as per the law of the country, child labor is strictly prohibited in the laws of Malaysia.
Human trafficking	No risk identified.	The plant will be operated as per the law of the country, human trafficking is strictly prohibited in the laws of Malaysia.

2.3.2 Human Rights

PAI1

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	PAI1 does not involve the encroachment of rights of IPs, LCs, and customary rights holders. The

	international human rights law and United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples are not violated.
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2.3.3 Indigenous Peoples and Cultural Heritage

PAI1

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	PAI1 does not have impact on Indigenous Peoples and their cultural heritage.

2.3.4 Property Rights

PAI1

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	PAI1 does not involve any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. The property rights are not held by any IPs, LCs, and customary rights holders, but the biogas powerplant owner.

2.3.5 Benefit Sharing

PAI1

PAI1 does not impact property rights as described in Section 2.3.4 above, hence this section is not applicable.

2.4 Ecosystem Health

PAI1

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified.	The project activity does not create any net negative environmental or socio-economic impacts. It involves capturing methane that would otherwise be released into the atmosphere and utilizing this captured methane to generate green electricity.
Soil degradation and soil erosion	No risk identified.	The project activity does not create any net negative environmental or socio-economic impacts. It involves capturing methane that would otherwise be released into the atmosphere and utilizing this captured methane to generate green electricity.
Water consumption and stress	No risk identified.	The project activity does not create any net negative environmental or socio-economic impacts. It involves capturing methane that would otherwise be released into the atmosphere and utilizing this captured methane to generate green electricity.

2.4.1 Rare, Threatened, and Endangered species

PAI1

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

PAI1

N/A

2.4.3 Ecosystem conversion

PAI1

N/A. Project is not an ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Grouped Project

The Grouped Project allows for the usage of the following methodologies, tools, standards and procedures.

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-I.D	Grid-connected renewable electricity generation	18.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system	07.0
Tool	TOOL 21	Demonstration of additionality of small-scale project activities	13.1
Tool	TOOL 32	Positive lists of technologies	4.0
Tool	-	General guidance on leakage in biomass project activities	3.0

PAI1

PAI1 involves the usage of the following methodologies, tools, standards and procedures.

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-I.D	Grid-connected renewable electricity generation	18.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system	07.0

Tool	TOOL 21	Demonstration of additionality of small-scale project activities	13.1
Tool	TOOL 32	Positive lists of technologies	4.0

3.2 Applicability of Methodology

PAI1

The scopes and applicability conditions for AMS-I. D is tabulated below:

Methodology ID	Applicability condition	Justification of compliance of PAI 1
AMS-I.D	As per paragraph 2 of AMS-I.D This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	Applicable. PAI1 involves renewable energy generation unit using biogas recovered from POME. The electricity generated will be supplied to the national grid.
AMS-I.D	As per paragraph 4 of AMS-I.D: This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s);	Applicable. PAI1 involves installation of a Greenfield biogas powerplant at the PAI1 location.

Methodology ID	Applicability condition	Justification of compliance of PAI 1
	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	
AMS-I.D	As per paragraph 5 of AMS-I.D Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	Not applicable. PAI1 does not involve hydro power plants.
AMS-I.D	As per paragraph 6 of AMS-I.D	Applicable.

Methodology ID	Applicability condition	Justification of compliance of PAI 1
	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The eligible limit of 15MW for renewable components are not exceeded. PAI1 powerplant has an installed capacity of 2.76 MW and a net export capacity of 2.00 MW, which is less than 15 kW.
AMS-I.D	As per paragraph 7 of AMS-I.D Combined heat and power (co-generation) systems are not eligible under this category.	Not Applicable PAI1 does not involve combined heat and power (co-generation) systems.
AMS-I.D	As per paragraph 8 of AMS-I.D In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not Applicable. PAI1 does not involve the capacity addition of renewable energy generation units.
AMS-I.D	As per paragraph 9 of AMS-I.D In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the	Not Applicable. PAI1 does not involve the retrofit, rehabilitation or replacement of powerplant.

Methodology ID	Applicability condition	Justification of compliance of PAI 1
	retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	
AMS-I.D	As per paragraph 10 of AMS-I.D In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	Not Applicable. PAI1 does not consider the emission reductions achieved under Type III, though applicable. This is explained in Section 1.1 and Section 1.15.
AMS-I.D	As per paragraph 11 of AMS-I.D In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Not Applicable. PAI1 will not be using biomass sourced from dedicated plantations which is plantation grown specifically for Project activity.

3.3 Project Boundary

Grouped Project

The following GHG sources, sinks, and reservoirs for the project and baseline scenarios should be assessed on relevance for additional project activity instances added to this Grouped Project:

Source		Gas	Included?	Justification/Explanation
Baseline	CO2 emissions from electricity generation in power plants that are displaced due to the project activity	CO ₂	Yes	Main Emission Source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO2 emissions from on-site consumption of fossil fuel due to project activity	CO ₂	Yes	Considered in the net power plant export capacity
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

PAI 1

The following boundaries apply to PAI1, as per AMS-I.D.

Source		Gas	Included?	Justification/Explanation
Baseline	CO2 emissions from electricity generation in power plants that are displaced due to the project activity	CO ₂	Yes	Main Emission Source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification
Project	CO2 emissions from on-site consumption of fossil fuel due to project activity	CO ₂	Yes	Considered in the net power plant export capacity
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
		Other	No	Excluded for simplification

The PAI1 boundary, physical locations of the various installations of technology taking place as part of the project activity based on the description provided in Section 1.12 and flows of mass and energy are indicated in the figure below.

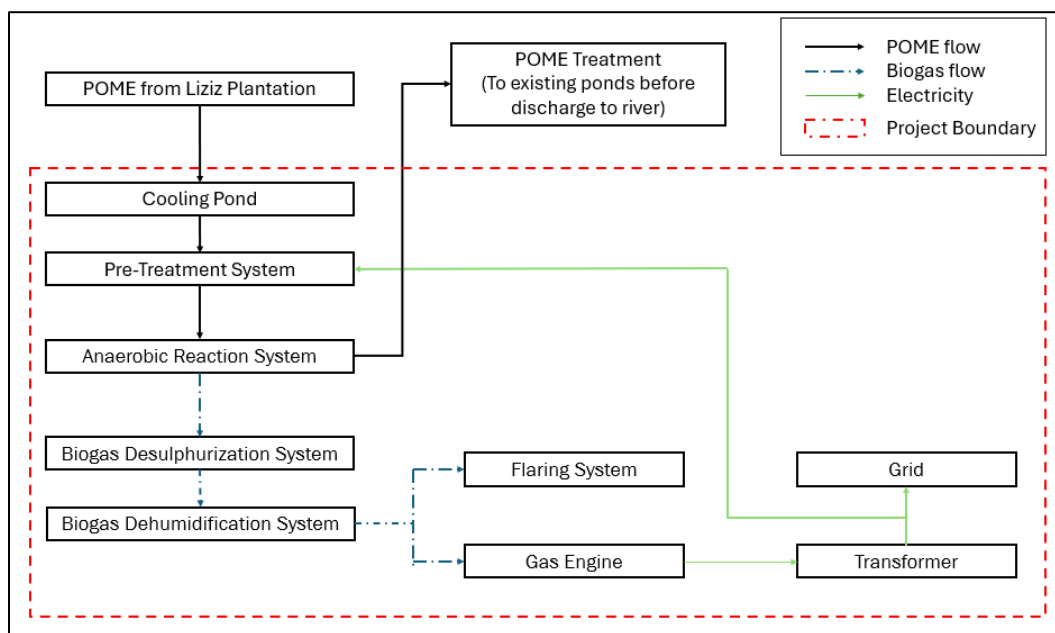


Figure 2: Project Boundary

3.4 Baseline Scenario

Grouped Project

As per AMS-I.D (Version 18.0), Chapter 5, the baseline methodology is set out. Since the Project is applying AMS-I. D as the sole methodology, the baseline scenario is derived based on it and described for each new inclusion of PAI.

PAI1

Chapter	Chapter Title	Paragraph	Applicability
5.2	Baseline scenario for Greenfield power plant	19	Yes
5.3	Baseline scenario for retrofit, rehabilitation or replacement	20	No
5.4	Baseline scenario for capacity addition	21	No

Given that the PAI1 involves a Greenfield power plant, according to AMS-I.D (Version 18.0), Paragraph 19, the baseline scenario is that the electricity supplied to the grid by the project, in this case, the Biogas Power Plant, would have been otherwise produced through the operation of existing grid-connected power plants and the incorporation of new generation sources into the grid. The quantification of baseline emissions is explained in detail in Section 5.1.

3.5 Additionality

Additionality for each PAI shall be demonstrated through the “General guidelines for SSC CDM methodologies, information on additionality” and “General guidance on leakage in biomass project activities”.

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

Not applicable. The Project is located in Malaysia, a Non-Annex 1 country and the project activities are not mandated by any law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Grouped Project

Each inclusion of PAIs shall demonstrate additionality by applying the “General guidelines for SSC CDM methodologies, information on additionality and “General guidance on leakage in biomass project activities”.

PAI 1

In Figure 1 of Tool 21: Demonstration of additionality of small-scale project activities (Version 13.1), the criteria for automatic additionality using provisions of small-scale (SSC) or microscale (MSC) additionality tools are listed. PAI1 is considered automatically additional based on the criteria as follows:

1. Is PA aggregate size $\leq$ SSC thresholds (15MW, 60GWh/y, 60ktCO₂e/y)?

Yes. The PA aggregate size is 2MW, 17.5 GWh/y and 13.7ktCO₂e/y.

2. Is PA/CPA comprised of one or more technologies from the positive list under TOOL32?

Yes. Based on Tool 32, Paragraph 12: The project activities and PoAs in an existing facility are deemed automatically additional if it is demonstrated that all of the following conditions are met

Requirements for additionality	Project Scenario
(a) The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation	Yes. The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the regulations of Malaysia.
(b) There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites	Yes. There is no regulation in Malaysia that requires the management of biogas from the project site, as discussed in Section 1.15.
(c) There is no capacity increase in the wastewater treatment system	Yes There has been no capacity increase in the wastewater treatment system
(d) No other alternative economic activity is expected to be undertaken on the land of the existing lagoon	Yes. No other alternative economic activity is expected to be undertaken on the land of the existing lagoon.

Requirements for additionality	Project Scenario
(e) The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW.	Yes. The biogas is used to generate electricity in a power plant and the total nameplate capacity is below 5MW.

PA1 meets all five (5) conditions listed above. As a result, PA1 is automatically additional.

3.6 Methodology Deviations

Not applicable as no deviations from the methodology were made.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

To be updated at the end of monitoring period.

5 Quantification Of Estimated GHG Emission Reductions and Removals

5.1 Baseline Emissions

(A) Grouped Project

The following formulae will be adhered by each PAIs in the Grouped Project to quantify the baseline scenario for grid connected renewable electricity generation. Based on the procedures set out in AMS-I.D, Version 18.0, 3 scenarios are outlined as follows:

1) **Baseline scenario for Greenfield power plant**

The baseline scenario is that the electricity delivered to the grid by the PAIs would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

2) **Baseline scenario for retrofit, rehabilitation or replacement**

For PAIs that involve retrofit, rehabilitation or replacement of an existing facility, the baseline scenario is the continuing operation of the existing plant. The methodology uses historical electricity generation data to determine the electricity generation of the existing plant in the baseline scenario, assuming that the historical situation observed prior to the implementation of the PAIs would continue. In the absence of the PAIs, the existing facility would continue to provide electricity to the grid at historical average levels until the time at which the electrical generation facility would be likely to be retrofitted, rehabilitated or replaced in the absence of the PAIs. From that point of time onwards, the baseline scenario is assumed to correspond to the project activity, and baseline electricity supply is assumed to equal the project's net electricity supply and no emission reductions are assumed to occur.

3) **Baseline scenario for capacity addition**

If the PAI is a capacity addition to existing grid-connected renewable energy power plant/unit, the baseline scenario is the existing facility that would continue to supply electricity to the grid at historical levels, until the time at which the generation facility would likely be replaced or retrofitted ($DATE_{BaselineRetrofit}$), and electricity delivered to the grid by the added capacity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. From that point of time onwards, the baseline scenario is assumed to correspond to the PAI, and no emission reductions are assumed to occur.

Baseline Emissions Calculation

As per AMS-I.D (Version 18.0), Chapter 5 sets out the baseline methodology for Grid Connected Renewable Electricity Generation. Since the Project is applying AMS-I.D as the sole methodology, the baseline emissions are calculated as follows:

Paragraph 22: Baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (1)}$$

Where:

BE_y	=	Baseline emissions in year y (tCO ₂)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO ₂ /MWh)

Paragraph 23: The emission factor shall be calculated in a transparent and conservative manner as follows:

- A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system”; or
- The weighted average emissions (in t CO₂/MWh) of the current generation mix. The data of the year in which project generation occurs must be used.

Paragraph 24: Calculations shall be based on data from an official source (where available) and made publicly available.

In this Grouped Project, the combined margin emission factor used is acquired directly from the Ministry of Energy, Science, Technology, Environment and Climate Change. The calculation⁸ is calculated according to the “Tool to calculate the emission factor for an electricity system”, estimating the OM, BM and CM for grid connected electricity system in Malaysia.

Based on the calculations, the OM emission factor based on fuel consumption for Peninsular Malaysia is 0.645 tCO₂/MWh while the BM emission factors based on fuel consumption for Peninsular Malaysia is 0.526 tCO₂/MWh. The default value used is 50% for OM and 50% for BM to calculate Combine Margin emission factor. This is due to the current situation where the thermal power plants is dominating in Peninsular Malaysia. As such, the combined margin emission factor for 2017, $EF_{grid,y}$ was calculated to be 0.585 tCO₂/MWh.

1) Baseline emissions for Greenfield power plant

Paragraph 26: If the project activity is the installation of a greenfield power plant, then:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad \text{Equation (2)}$$

Where:

$$EG_{PJ,facility,y} = \text{Quantity of net electricity generation supplied by the project plant/unit to the grid in year } y \text{ (MWh)}$$

2) Baseline emissions for Capacity addition

Since the Grouped Project involves biogas and biomass powerplants, the capacity addition components for wind, solar, wave, tidal, hydro and geothermal powerplants are not applicable.

Paragraph 29: For project activities that involve the addition of renewable energy generation units at an existing biomass based power generation facility, where the existing and new units share the use of common and limited renewable resources (e.g. biomass residues), the potential for the project activity to reduce the amount of renewable resource available to, and thus electricity generation by, existing units must be considered in the determination of baseline emissions, project

⁸ <https://www.mgtc.gov.my/wp-content/uploads/2019/12/2017-CDM-Electricity-Baseline-Final-Report-Publication-Version.pdf>

emissions, and/or leakage, as relevant. The baseline energy generation ($EG_{PJ,y}$) corresponding to the net increase in electricity production associated with the project should be calculated as follows:

$$EG_{PJ,y} = \begin{cases} (EG_{PJ,facility,y} - EG_{BL,existing,y}), & \text{until } DATE_{BaselineRetrofit} \\ 0, & \text{after } DATE_{BaselineRetrofit} \end{cases} \quad \text{Equation (3)}$$

Where:

$$EG_{BL,existing,y} = MAX(EG_{actual,y}, EG_{estimated,y}) \quad \text{Equation (4)}$$

Where:

- $EG_{BL,existing,y}$ = The net electrical energy that would have been supplied by the plant/unit to the grid in the absence of the project activity determined as maximum of $EG_{actual,y}$ and $EG_{estimated,y}$ in year y (MWh)
- $EG_{actual,y}$ = The net electrical energy produced and supplied to the grid by existing power plant/unit (installed before the project activity) in year y in the absence of the project activity (MWh)
- $EG_{estimated,y}$ = Estimated net electrical energy that would have been produced by the existing units under the observed availability of the renewable resource in year y (MWh)
- $DATE_{BaselineRetrofit}$ = Point in time when the existing equipment would need to be replaced in the absence of the project activity (date)

Paragraph 30: If the existing units shut down, are derated, or otherwise become limited in production, the project activity should not get credit for generating electricity from the same renewable resources that would have otherwise been used by the existing units (or their replacements). Therefore, the equation for $EG_{BL,existing,y}$ still holds, and the value for $EG_{estimated,y}$ should continue to be estimated assuming the capacity and operating parameters are the same as that at the time of the start of the project activity.

3) Baseline emissions for retrofit rehabilitation or replacement

Paragraph 32: The baseline energy generation ($EG_{PJ,y}$) corresponds to the net increase in electricity production associated with the project is calculated as follows:

$$EG_{PJ,y} = \begin{cases} (EG_{PJ,facility,y} - EG_{BL,retrofit,y}), & \text{until } DATE_{BaselineRetrofit} \\ 0, & \text{after } DATE_{BaselineRetrofit} \end{cases} \quad \text{Equation (5)}$$

$$EG_{BL,retrofit,y} = MAX(EG_{historical}, EG_{estimated,y}) \quad \text{Equation (6)}$$

$EG_{BL,retrofit,y}$ = The net electrical energy that would have been supplied by the plant/unit to the grid in the absence of the project activity determined as maximum of $EG_{historical,y}$ and $EG_{estimated,y}$ in year y (MWh)

Determination of $EG_{historical}$

Paragraph 33: Average of historical net electrical energy levels delivered by the existing facility, spanning all data from the most recent available year (or month, week or other time period) to the time at which the facility was constructed, retrofit, or modified in a manner that significantly affected output (i.e. by 5 per cent or more), shall be used.

Paragraph 34: To determine $EG_{historical}$, PAIs may choose between two historical periods. This allows some flexibility: the use of the longer time period may result in a lower standard deviation and the use of the shorter period may allow a better reflection of the (technical) circumstances observed during the more recent years.

Paragraph 35: Each PAIs may choose among the following two time spans of historical data to determine $EG_{historical}$:

- (c) The three last calendar years (in case of hydro power plants five years)⁹ prior to the implementation of the project activity; or
- (d) The time period from the calendar year following $DATE_{hist}$, up to the last calendar year prior to the implementation of the project, as long as this time span includes at least three calendar years (in case of hydro power plants five years), where $DATE_{hist}$ is latest point in time between:
 - (i) The commissioning of the plant/unit;
 - (ii) If applicable: the last capacity addition to the plant/unit; or

⁹ In the case that three years of historical data are not available due to recent retrofits or unusual circumstances such as natural disasters, conflicts, and transmission constraints, a new methodology or methodology revision shall be proposed.

- (iii) If applicable: the last retrofit or rehabilitation of the plant/unit.

Paragraph 36: In case of rehabilitation where the power plant/unit did not operate for last three (in case of hydro power plants five years) calendar years before the rehabilitation starts, $EG_{historical}$ is equal to zero.

Calculation of $DATE_{BaselineRetrofit}$

Paragraph 37: In order to estimate the point in time when the existing equipment would need to be replaced/retrofitted in the absence of the project activity ($DATE_{BaselineRetrofit}$), project participants may take into account the typical average technical lifetime of the type equipment, which shall be determined and documented as per the “Tool to determine the remaining lifetime of equipment”.

Paragraph 38: The point in time when the existing equipment would need to be replaced/retrofitted in the absence of the project activity should be chosen in a conservative manner that is, if a range is identified, the earliest date should be chosen.

For each PAI, detailed quantification of estimated GHG emission reductions and removals shall be detailed accordingly based on the scenario for each PAI.

(B) PAI1

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (1)}$$

Where:

BE_y	=	Baseline emissions in year y (tCO ₂)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO ₂ /MWh)

PAI1 involves a Greenfield biogas powerplant located in Malaysia. The combined margin emission factor, $EF_{grid,y}$ of 0.585 tCO₂/MWh is used, as detailed above.

Since PAI1 involves the installation of a greenfield power plant, then:

$EG_{PJ,y} = EG_{PJ, facility,y} \cdot EG_{PJ,y}$, the estimated quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of PAI1 is estimated based on the Final Declared Annual Availability Document released by the Sustainable Energy Development Authority Malaysia (SEDA Malaysia) and summarised in the table below.

Crediting Period, Year Y	$EG_{PJ,y}$ (MWh)	$EF_{grid,y}$ (tCO ₂ e/ MWh)	BE_y (tCO ₂ e)
26-July-2023 to 31-December-2023	7632	0.585	4464.72
01-January-2024 to 31-December-2024	17520	0.585	10249.20
01-January-2025 to 31-December-2025	17520	0.585	10249.20
01-January-2026 to 31-December-2026	17520	0.585	10249.20
01-January-2027 to 31-December-2027	17520	0.585	10249.20
01-January-2028 to 31-December-2028	17520	0.585	10249.20
01-January-2029 to 31-December-2029	17520	0.585	10249.20
01-January-2030 to 26-July -2030	17520	0.585	5784.48

5.2 Project Emissions

Grouped Project

The procedure for quantification of project emissions is outlined in AMS-I.D (Version 18.0), Paragraph 39 to 41.

As per Paragraph 39, for most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in the most recent version of “ACM0002: Grid-connected electricity generation from renewable sources”:

- a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- b) Emissions from water reservoirs of hydro power plants.

As per Paragraph 40, CO₂ emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the latest version of the “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”.

As per Paragraph 41, in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used.

PAI1

PAI1 is not included in the categories of project activities listed in Paragraph 39, hence $PE_y = 0$.

PAI1 does not produce CO₂ emissions from on-site consumptions of fossil fuels; hence paragraph 40 is not applicable.

PAI1 does not source biomass from dedicated plantations (plantations which are plantations grown specifically for Project activity), hence paragraph 41 is not applicable.

5.3 Leakage Emissions

Grouped Project

As per AMS-I. D (Version 18.0) Paragraph 42, general guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues.

PAI1

Since PAI1 is a biogas project, the leakage emissions, $LE_y = 0$.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Grouped Project

As per AMS-I. D (Version 18.0) Paragraph 43:

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y \quad \text{Equation (7)}$$

Where:

ER_y	=	Emission reductions in year y (t CO ₂)
BE_y	=	Baseline Emissions in year y (t CO ₂)
PE_y	=	Project emissions in year y (t CO ₂)
LE_y	=	Leakage emissions in year y (t CO ₂)

PAI1

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
26-July-2023 to 31-December-2023	4464.72	0	0	4464.72	0	4464.72
01-January-2024 to 31-December-2024	10249.20	0	0	10249.20	0	10249.20
01-January-2025 to 31-	10249.20	0	0	10249.20	0	10249.20

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated removal VCU's (tCO ₂ e)	Estimated total VCU's (tCO ₂ e)
December-2025						
01-January-2026 to 31-December-2026	10249.20	0	0	10249.20	0	10249.20
01-January-2027 to 31-December-2027	10249.20	0	0	10249.20	0	10249.20
01-January-2028 to 31-December-2028	10249.20	0	0	10249.20	0	10249.20
01-January-2029 to 31-December-2029	10249.20	0	0	10249.20	0	10249.20
01-January-2030 to 26-July -2030	5784.48	0	0	5784.48	0	5784.48
Total	71744.40	0	0	71744.40	0	71744.40

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,y}
Data unit	tCO ₂ / MWh
Description	Grid Emission Factor for Peninsula Malaysia
Source of data	Ministry of Energy, Science, Technology, Environment and Climate Change, Malaysia
Value applied	0.585
Justification of choice of data or description of measurement methods and procedures applied	The calculation¹⁰ is calculated by Malaysian Green Technology Corporation in the 2017 CDM Electricity Baseline for Malaysia according to the “Tool to calculate the emission factor for an electricity system”, estimating the OM, BM and CM for grid connected electricity system in Malaysia. Based on the calculations, the OM emission factor based on fuel consumption for Peninsular Malaysia is 0.645 tCO₂/MWh while the BM emission factors based on fuel consumption for Peninsular Malaysia is 0.526 tCO₂/MWh. The default value used is 50% for OM and 50% for BM to calculate Combine Margin emission factor. This is due to the current situation where the thermal power plants is dominating in Peninsular Malaysia. As such, the combined margin emission factor for 2017, EF_{grid,y} was calculated to be 0.585 tCO₂/MWh
Purpose of data	Calculation of baseline emissions
Comments	The above study conducted in 2017 will be updated with the most recent figures if the Ministry of Energy, Science, Technology, Environment and Climate Change in Malaysia publishes newer studies. Accordingly, the Grid Emission Factor will be revised to reflect the latest data.

¹⁰ <https://www.mgtc.gov.my/wp-content/uploads/2019/12/2017-CDM-Electricity-Baseline-Final-Report-Publication-Version.pdf>

6.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	On site measurement
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter
Frequency of monitoring/recording	Continuous monitoring, hourly measurement and at least monthly recording
Value applied	Values to be monitored ex-post
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	Maintenance and calibration of the meters will be done according to the national grid operator, Tenaga National Berhad's requirement and standard operating procedures. The data obtained from the meter can be cross-checked against the electricity sales invoices from the national grid operator.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

6.3 Monitoring Plan

PAI1

Based on the monitored data and parameters set out in Section 5.2, the monitoring plan on process and schedule for obtaining, recording, compiling and analysing the monitored data is developed. From Section 5.2, the monitored parameter is $EG_{PJ,y}$, which is the quantity of net electricity generation supplied by the project to the grid in year y.

Additionally, for enhanced safety, a gas tester operated by a service engineer manages any biogas leaks, ensuring excess gas not used in electricity generation is burned off safely in an auto-activated flaring system triggered by specific pressure levels. The service engineer also monitors all vital biogas operation and management parameters through a comprehensive integrated system.

The monitoring plan presented in this PDD assures that real, measurable, long-term GHG emissions reductions are monitored, recorded and reported. This plays an important role in the final VCU's of the project. The monitoring plan will be executed by project proponents along with project implementation partner throughout the project crediting period, with details specified as follows:

1. Monitoring structure

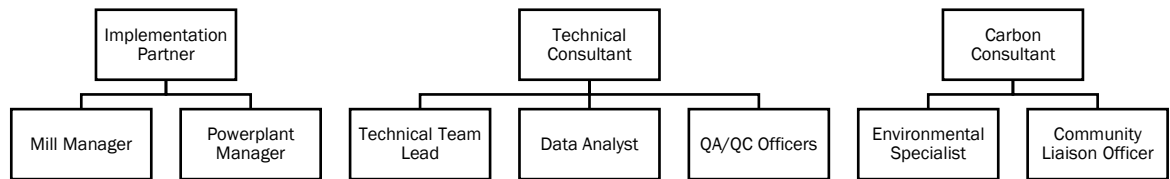


Figure 3: Project VCS Organizational Chart

Personnel	Description of Roles in Monitoring Plan
Implementation Partner	- Oversees the project implementation. - Ensuring that all activities align with the project's goals and standards.
Mill Manger	- Manages the day-to-day operations of the palm oil mill, including oversight of POME generation and initial handling. - Ensures consistency of feedstock supply
Powerplant Manager	- Responsible for the biogas power plant operations - Ensuring optimal electricity generation from the captured biogas. - Reports on the electricity generation from the powerplant to relevant authority and project director. - Direct service engineers on the management of leakage of biogas.
Technical Consultant	- Provides expert advice on the technical aspects of the project, including the design and optimization of the biogas capture and electricity generation process. - Monitoring operation, maintenance, and calibration of powerplant equipment including monitoring equipment.
Technical Team Lead	- Manages the technical team. - Ensuring the proper operation and maintenance of the biogas system and monitoring equipment.
Data Analyst	Aggregates, analyzes, and reports on the data collected from the monitoring equipment to assess the performance of the biogas power plant.
QA/QC Officers	- Implement quality assurance and control measures. - Verifying the accuracy and reliability of the monitoring data.

Personnel	Description of Roles in Monitoring Plan
Carbon Consultant	- Ensuring that the monitoring plan adheres to protocols required by carbon standards, i.e. Verra standards. - Advising on the documentation and verification process needed for the sale and transfer of carbon credits. - Guiding the project through the validation and verification processes by a third-party auditor.
Environmental Specialist	- Evaluates the project's environmental impact. - Ensuring compliance with environmental regulations and sustainability standards
Community Liaison Officer	Serves as the bridge between the project and the local community, addressing any concerns and ensuring the project's activities are communicated transparently

2. Measurement and Recording

Precise metering equipment connected to the power generation system is used to measure the quantity of electricity generated and supplied to the grid. The equipment is ensured to be calibrated according to the National Grid Operator's instructions to maintain accuracy. The meter data is also cross checked with the electricity invoice from the grid operator.

3. Data storage and aggregation

Digital data management system for storing, aggregating and collating electricity generation data is formed. This is to allow for secure data logging and easy retrieval for analysis and reporting.

4. Quality Assurance and Quality Control (QA/QC)

All metering equipment for monitoring will be chosen and calibrated regularly for accuracy by qualified parties according to the National Grid Operator, TNB.

5. Internal auditing

Internal auditing procedures are developed to regularly review the monitoring process and ensure compliance with the validated monitoring plan.

6. Non-conformance Handling

Protocols for identifying, reporting, and addressing non-conformances or deviations from the validated monitoring plan are developed, which includes corrective action plans and preventive measures.

7. Sampling approaches used

Not applicable, as the parameters collected are continuous parameters.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

PAI1

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Value applied:	Refer to Section 7.2 for the actual electricity generation data, which totals 3574.2 MWh for the period from July 23, 2023, to July 31, 2024.
Comments	NA

7.2 Baseline Emissions

PAI1

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (1)}$$

Where:

- BE_y = Baseline emissions in year y (tCO₂)
- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
- $EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

$EG_{PJ,y}$ is the sum of actual electricity supplied to the grid each month (kWh) from 23 July 2023 to 31 July 2024 as detailed in the table below. Meanwhile, $EF_{grid,y}$ is taken to be 0.585 as per described in Section 5.1.

Year	Actual Electricity Generation (kWh)	Actual Electricity Generation (MWh)	Baseline Emission (tCO ₂ e)
23-31 July 2023	108843.0	108.8	63.7
Aug 2023	336026.0	336.0	196.6
Sep 2023	117133.0	117.1	68.5
Oct 2023	0.0	0.0	0.0
Nov 2023	117230.0	117.2	68.6
Dec 2023	219597.0	219.6	128.5
Jan 2024	633251.0	633.3	370.5
Feb 2024	456574.0	456.6	267.1
Mar 2024	530148.0	530.1	310.1
April 2024	26796.0	26.8	15.7
May 2024	1578.0	1.6	0.9
June 2024	295307.0	295.3	172.8
July 2024	731745.0	731.7	428.1
$EG_{PJ,y}$ (23 July 2023- 31 July 2024)		3574.2	2090.9

7.3 Project Emissions

PAI1

PAI1 is not included in the categories of project activities listed in Paragraph 39, hence $PE_y = 0$.

PAI1 does not produce CO₂ emissions from on-site consumptions of fossil fuels; hence paragraph 40 is not applicable.

PAI1 does not source biomass from dedicated plantations (plantations which are plantations grown specifically for Project activity), hence paragraph 41 is not applicable.

7.4 Leakage Emissions

As per AMS-I. D (Version 18.0) Paragraph 42, general guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues.

PAI1

Since PAI1 is a biogas project, the leakage emissions, $LE_y = 0$.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

As per AMS-I. D (Version 18.0) Paragraph 43:

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y \quad \text{Equation (8)}$$

Where:

ER_y	=	Emission reductions in year y (t CO ₂)
BE_y	=	Baseline Emissions in year y (t CO ₂)
PE_y	=	Project emissions in year y (t CO ₂)
LE_y	=	Leakage emissions in year y (t CO ₂)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
23 July to 31-Dec-2023	525.8	0	0	525.8	0	525.8
01-Jan-2024 to 31-July-2024	1565.1	0	0	1565.1	0	1565.1
Total	2090.9	0	0	2090.9	0	2090.9

The table below summarizes the ex-ante estimated reductions and the achieved reductions for PAI1, along with an explanation of the differences between the two.

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference	
23 July to 31-Dec-2023	4464.7	525.8	88.2	1.	Insufficient availability of Palm Oil Mill Effluent (POME) feedstock due to lower production rates at the mill.
01-Jan-2024 to 31-July-2024	5978.7	1565.1	73.8		
Total	10443.42	2090.9	80.0	2.	Extensive maintenance activities on electrical systems during the initial year of operation, impacting overall efficiency.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.